PROGRAMME STATUS · CONFIDENTIAL

GRX930 on wafer.space

Status and indicative schedule — 26 October 2026, 20:10 local (27 October 2026, 00:10 UTC). Every figure is measured from a run or a report in the tree; the routing number is live and moves while the current pass runs.

75,455	258,713	1,710	12–50 min
DRC violations at end of routing pass 0, from 461,079	live count, pass 1, falling — deemed converging	antenna diodes inserted by the flow, which now completes	cost of one routing round, against 16–27 hours before

IN ONE PARAGRAPH

The silicon-side design is done and verified in simulation; the physical implementation is one step from done. The part intended for the shuttle is a **single-core** GRX930 test chip in the template's 1×1 pad ring, booting from its own ROM with no external loader. The flow now completes synthesis, placement, clock tree insertion, global routing and — for the first time — **the full antenna repair, in-flow, with 1,710 diodes inserted**, and it is inside detailed routing with a curve that is falling: 461,079 → 75,455 violations in the first pass, and 542,150 → 258,713 in the second after the repair re-routed the diode-carrying database. Iterations now cost 12–50 minutes rather than 16–27 hours. It is not yet clean, and the pass-over-pass restart level is what decides the date. On the shuttle side the position is unchanged and now verified: **Run 3's full-size slots are sold out**, so the binding deadline is whichever date Run 4 is announced with.

What changed since the 24 October edition

- **Antenna repair is closed** — the step that aborted (`GRT-0183`) now completes in-flow, 1,710 diodes inserted, no jumper-only fallback.
- **Detailed routing turned** — the last edition's 1.03 M → 1.65 M "each pass worse" is now **75,455** at the first pass end, converging in the second.
- **Iteration cost collapsed** — 16–27 hours per round became 12–50 minutes, and a guard stops any round that overruns.

WHAT IS ACTUALLY GOING TO THE SHUTTLE

Process	GF180MCU, 180 nm, 5LM_1TM_11K, `gf180mcu_fd_sc_mcu7t5v0`
Slot / die	1×1, **3932 × 5122 µm**, origin (0,0), dbu 0.001 µm
Core	3048 × 4238 µm = **12.93 mm²**, ~46 % utilisation
CPU	1× RV64IMAC, 5-stage, 512 B I + 512 B D cache (16 × 32 B lines)
Memory	2 KB on-chip SRAM + 1 KB boot ROM at `0x0001_0000` (`CORE_RESET_PC`)
I/O	UART, GPIO, one external interrupt, 74 pad openings (template ring)
Frequency	see *Timing* below — the 40 ns data paths close (25 MHz) once the reset tree is fixed; 50 MHz needs a re-floorplan

On "quad-core"

The quad-core SoC (4× RV64IMAC, 2 NPUs, 64 KB shared L2, write-through MSI-lite directory) is a real, FPGA-validated design — timing closed on the Artix-7 at ~121 MHz, 0 failing endpoints — but it is not this shuttle part. Its memory architecture was frozen on 28 Oct 2026 as **configuration B**: main memory moves off-chip, removing 26.8 mm² of SRAM macro and putting the part at **86.4 mm² of content** (46.2 mm² of cells + 40.2 mm² of macro) — **not the ~46 mm² earlier editions quoted**, which counted the cells and none of the memories. On the area model that costed the 2×4, that is a **~165 mm² die at 46 % row utilisation — a ~12.9 mm square**, and **≈€151k** at EUROPRACTICE's published €913/mm² (≈€147k at 161 mm², ≈€161k at a 177 mm² relief valve). GlobalFoundries' own shuttle terms are unpublished, and **wafer.space cannot build it:** 20.12 mm² is its largest die (confirmed 27 Oct 2026). Putting it on silicon is therefore its own decision with its own supplier, downstream of this vehicle proving the flow. Details: `wafer_space_integration/quadcore_submission_scope.md` §4.

STATUS BY STAGE

STAGE	STATE	EVIDENCE
RTL, boot, bring-up	done	boots from ROM with empty SRAM (`+NO_PRELOAD` PASS); `-DROM_DISABLE` control fails; cache index/tag and MMIO read-hold fixes landed
Pad ring + adapter	done	template `chip_top.sv` / `slot_1x1.yaml` unmodified; adapter lint PASS (caught two real defects); precheck stage-3 slot size passes
Global routing	done	`GRT-0183` root-caused to a no-path in a bounded search region; `GRT_ADJUSTMENT 0.1` converges with 0 overflow on every layer in 8 sweeps
Antenna repair	done	in-flow on the 26Q4 engine: **1,710 diodes inserted**, step completes; the same step aborted on the pinned build
Active-layer density	done, not yet on this core	replacement fill reaches COMP 31.27 % / Poly2 26.25 % (floors 25 / 14) with 0 deck violations; the PDK default missed both
GDS stream-out	done	rebuilt from DEF + PDK library; the previous artifact had LEF-indexed layers, 229 cell abstracts and 30,130 cells in the wrong rows
Netlist vs layout (structural)	done	CDL-vs-DEF compare `MATCH`: 53,745 instances, 0 unmatched, 0 DEF-only
Detailed routing	in progress — falling	pass 0 **461,079** → **75,455**; repair re-route **542,150** → **258,713** and iterating; gate requires 0
Timing	reset tree	data paths close at 40 ns (−0.064 ns on the pinned engine); a 2,132-load `clkbuf_16` on reset does not. Hold −2.44 ns on both engines
Precheck on a core GDS	never run	pad-mask offset (74 openings 0.5 µm inside the golden mask) and a 5 nm grid snap are known items
Shuttle slot	unavailable	Run 3 full-size slots sold out (confirmed 26 Oct); Run 4 undated

THE BLOCKER, IN NUMBERS

The live arm is the die with its antenna repair completed, routed on OpenROAD 26Q4, capped at 8 optimisation rounds per pass and re-routed up to 10 times by the antenna-repair loop. Two passes are in the log:

PASS	START	ROUNDS, IN ORDER	STANDING
0	461,079	461,079 → 273,018 → 251,227 → 145,595 → 115,794 → 100,318 → 89,865 → 83,371 → **75,455**	capped after 8 rounds, −83.6 %
1	542,150	542,150 → 297,665 → 275,717 → **258,713** → (4th round open)	converging, mean gain 20.7 %/round

The restart is higher than the pass-0 floor because the antenna repair inserted its diodes into the routed database and the router re-lands every net they touch. The question the schedule turns on is not the within-pass slope — that is healthy — but whether the **next** pass starts below 542,150 and ends below 75,455. If the restart level falls pass over pass, the loop closes; if it stays above the previous floor, the repair is adding more disturbance than the router removes and the configuration changes below are back on the critical path.

Three things changed the cost of finding out. Iterations are measured at **12–50 minutes**, with a single 84-minute outlier, against the 16–27 hours recorded on the earlier core-only curve. A wall-clock and regression guard watches every pass and stops the first of: a round past 2 h, a pass past 8 h, 30 minutes of silence in an open round, a count 25 % above the pass's best, or three rounds each buying less than 0.25 %. It fired once today (a 2 h round, on an earlier launch) and the arm was relaunched; since then every round has landed inside its budget. And the step log is line-buffered end to end, so a silent router and a stalled one are now distinguishable — the guard's 30-minute rule is real rather than nominal.

The one standing operational risk is disk: the flow box is at **9.4 GB free of 150 GB (94 % used)**, and a pass writes a ~400 MB database. A guard polls the volume and prunes old route trees at the soft threshold, so the arm is protected, but headroom is thin

and the spare ARM box (217 GB free) is where the next arm would go.

TIMING

The 24 October edition carried two STA readings of the same database and could not reconcile them. They are reconciled now, and neither was a design change: the −345 ns row was taken *before* the post-CTS resizer, and the flow's own next STA step reads −0.100 ns on that same database. Re-running the post-repair STA on the pinned engine, through the flow's own tool, reproduces its control to five figures:

MEASUREMENT	ENGINE	SETUP WNS	SETUP TNS	ENDPOINTS	40 NS?
post-CTS, before the resizer (13 Oct)	pinned	−345.20 ns	—	30,490	superseded
post-resizer (13 Oct)	pinned	−0.100 ns	−2.01 ns	43	yes
post-repair (27 Oct)	pinned	**−0.064 ns**	−0.62 ns	28	yes
post-repair (27 Oct)	26Q4	−8.646 ns	−6,808 ns	2,393	no

The two engines disagree by 8.6 ns on identical input, and the whole difference is one net: `split25`, a `clkbuf_16` driving **2,132 loads** with a 21–47 ns transition against a 3 ns limit. Both engines flag that transition as illegal and charge it differently (0.39 ns against 18.55 ns at the load pin). So the useful statement is not a single WNS:

- the **40 ns data paths close** (−0.064 ns, i.e. 25 MHz), and did before the repair;
- the **reset distribution does not**, on either engine, and that is a buffering fix rather than a frequency decision;
- hold is −2.44 ns on both engines, so it is a real item for the resizer.

Until the reset tree is fixed, no frequency should be quoted; after it, 40 ns is the target and 50 MHz needs a re-floorplan.

INDICATIVE TIMEFRAMES

MILESTONE	ESTIMATE	CONFIDENCE	WHAT MOVES IT
A routing curve that reaches zero	1–3 weeks of iteration	medium	whether the pass-over-pass restart level falls; a configuration change is ~2 h per experiment in the fast loop
Clean route → GDS → precheck pass	2–4 weeks after that	medium-low	each pre-routing change forces a re-route; each precheck run is ~9 h, and the pad-mask/grid items need a re-spin
GDS ready to upload	**late November 2026 at the earliest; December 2026–Q1 2027 is the realistic band**	medium-low	routing convergence above; nothing else is on the critical path
Run 3 (clean GDS 16 Dec 2026)	**not achievable, and moot**	high	full-size slots sold out; purchase closed to us
Run 4	date unknown	—	wafer.space has not published one; we will be precheck-clean and uploadable on day one
Silicon in hand	follows the shuttle, not us	high	Run 3 parts are quoted for Q2 2027; Run 4 ships on its own schedule

Budget unchanged: **$8,000** for the 1×1 slot, **$1,500** for chip-on-board, with the CoB option kept alive in the layout (the four marker cells are already instantiated).

WHAT WE NEED DECIDED

1. **Reset buffering, then the target clock** — fix the 2,132-load reset buffer so both engines agree, and confirm 40 ns (25 MHz) as the target; 50 MHz needs a re-floorplan. The part boots either way.
2. **Quad-core vehicle** — whether the four-core SoC should be scoped as a larger-slot submission in parallel, or held until this vehicle has taped out.
3. **Slot strategy** — watch for a released Run 3 full-size slot and subscribe for Run 4 notification, or accept that the first silicon lands on Run 4's calendar.

EVIDENCE TRAIL

Every figure above traces to a run or a report. These are the ones a reviewer would ask for.

CLAIM	SOURCE	HOW IT WAS MEASURED
Global routing closes at 0 overflow	run `ws_v3` step 37	one-knob probes on the frozen pre-GRT state, ≈7 min each; `wafer_space_integration/probe_grt.sh`
Antenna repair completes, 1,710 diodes	run `dielb01` step 04	the 26Q4 engine in-flow; the pinned build aborts with `GRT-0183` on the same database
DRT pass 0: 461,079 → 75,455	run `dielb01` step 06	the step's own `DRT-0195` / `DRT-0199` lines, segmented per pass by `watch_drt_curve.sh`
Timing: −0.064 ns pinned, −8.646 ns 26Q4	`sta_engine_reconciliation.md`	single-step LibreLane resume on identical state; the control reproduces the flow's −0.0999734 ns to five figures
Density 31.27 % / 26.25 %, 0 deck violations	`wafer_space_integration/fill/`	the gf180mcuD deck in the pre-check image, deep mode, on the flow's own GDS
Structural LVS `MATCH`	`build/lvs_chip/net_compare.json`	CDL vs DEF: 53,745 instances each, 0 unmatched, 0 DEF-only
Run 3 full-size slots sold out	Crowd Supply campaign page	read 26 Oct 2026; the site still shows 9 Dec purchase / 16 Dec GDS dates

Prepared from the run tree on the flow box and the reports in `Claude outputs/` (`ws_run3_readiness_gaps.md` is the full gap register). Figures are point-in-time; the detailed routing count changes with every round.